

Duality Theory

Let $f: X \times Y \rightarrow \mathbb{R}$. We define

$$\sup(x) := \sup_{y \in Y} f(x, y)$$

$$\inf(y) := \inf_{x \in X} f(x, y)$$

together with their associated problems, i.e.

(p)	$\inf_x \sup(x)$	primal
(d)	$\sup_y \inf(y)$	dual.

Weak duality theorem:

$$\underbrace{\sup_{y \in Y} \inf(y)}_{=: d^*} \leq \underbrace{\inf_{x \in X} \sup(x)}_{=: p^*}$$